EE558 –Optical Fiber Comm. Systems
Prof. Alan Willner

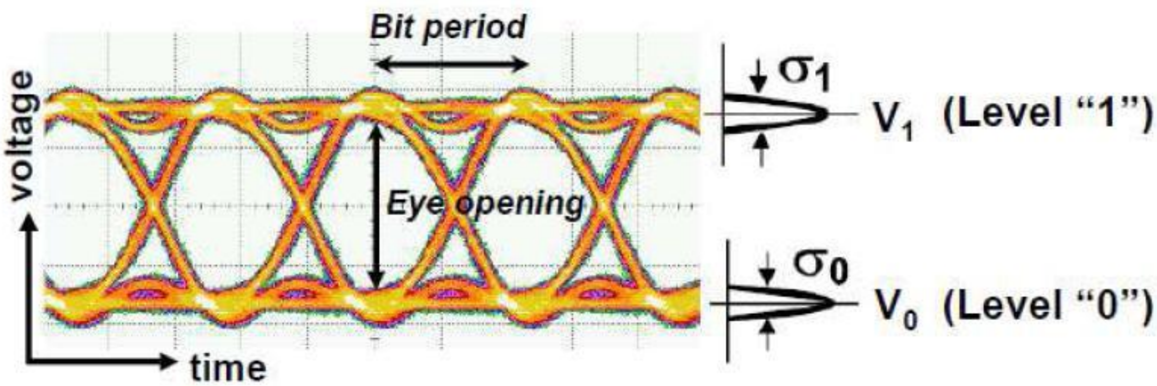
EE558 HW5

Problem 1:

(a) Draw the data constellation for QPSK, BPSK, and ASK.

(b) What is the spectral efficiency for QPSK.

Problem 2:



Q factor (i.e., signal quality factor) can be evaluated from the measured eye diagram, by evaluating the mean and the variance of the signal trances in the upper and lower parts of the eye diagram. Q factor is directly related to the bit-error rate (BER): BER $\approx 0.5 \times \mathrm{erfc}(Q/\sqrt{2})$.Calculate the Q factors that correspond to BER = 10^{-9} and 10^{-3} (Hint: You may want to find the definition of the function erfc(), google it.)

Problem 3:

Specifically, Q factor could be defined as $Q = (V_1 - V_0)/(\boldsymbol{\sigma}_1 + \boldsymbol{\sigma}_0)$, where V_1, V_0, $\boldsymbol{\sigma}_1$ and $\boldsymbol{\sigma}_0$ are defined as shown in the Figure of Problem 2. In a binary circuit, the RMS noise is 50 nV for both "0" and "1" levels (i.e., $\boldsymbol{\sigma}_1$ and $\boldsymbol{\sigma}_0$). The signal voltages are V_1 and V_0. Assuming V_0 is 0 and the threshold voltage is $V_1/2$. Find V_1 in order to make the average probability of error = 10^{-9}. (Note that in this case, because the Gaussian curves are symmetrical about the threshold voltage, the individual probabilities of error are equal. You may use your results in Problem 2 to answer this question.)

Problem 4:

Describe briefly how an eye diagram changes when you increase and decrease the cutoff frequency of a lowpass filter following the direct detection of an ASK NRZ signal.

Problem 5:

BER for different modulation schemes (i.e., amplitude-shift-keying and phase-shift-keying) are given below:

ASK (Homodyne): BER = $\frac{1}{2}\left[erfc\left(\sqrt{\frac{\eta N_p}{2}}\right)\right]$

ASK (Heterodyne): BER = $\frac{1}{2}\left[erfc\left(\sqrt{\frac{\eta N_p}{4}}\right)\right]$

PSK (Homodyne): BER = $\frac{1}{2}\left[erfc\left(\sqrt{2\eta N_p}\right)\right]$

PSK (Heterodyne): BER = $\frac{1}{2}\left[erfc\left(\sqrt{\eta N_p}\right)\right]$

N_p is the number of photons per bit. Which scheme provides the most sensitivity at BER of 10^{-9}? Why?